

# From cohort to counterfactuals

How the CatBoost model is trained, tuned, evaluated and explained · 187 patients · random seed 42

■ used to fit

■ held out within training

■ test set (held out)

■ search / decision

## Which patients are held out

Repeated Stratified K-Fold: 4 folds × 10 repeats = 40 train/test splits



## Which training patients tune the model, and which set its threshold

Stratified K-Fold ×2 · the training set is split 3-fold twice, with different seeds



### Optuna TPE — 100 trials per train/test split

Objective: mean AUPRC across the 3 tuning folds

Tunes: depth · learning rate · L2 leaf regularization · positive-class weight

carried forward: winning hyperparameters + the 3 threshold folds

## Refit, threshold selection, and test-set evaluation

· per train/test split

### 1 · Refit

Winning params refit on the full training set

### 2 · Out-of-fold scores

One held-out score per patient

### 3 · Decision threshold

Maximize F1 on the OOF curve

### 4 · Test-set evaluation

Refit model + its threshold scored once → performance metrics and plots

## Explaining the model

· repeat 1 only: its 4 fold models and thresholds, saved and reused

### Feature importance & SHAP

· the 4 fold models, refit

Refit with identical params, data and seed — reproduces the stored models

#### Feature Importance

get feature importance per fold model → 4-model heatmap

#### SHAP

Explainer over each fold's own test set → mean |SHAP| heatmap

### Counterfactuals (DiCE)

· the same 4 models, used as stored — no refit

Each wrapped with its own tuned threshold, so DiCE decides at that cut-off, not 0.5

#### Instances of interest

misclassified, or  
 $|p - \text{threshold}| \leq 0.08$

#### DiCE genetic search

varies only the 6 actionable features

#### Outputs per fold

local CFs + global CF importance